

Knowledge Distillation-Aided Deep Unfolding for Hybrid Beamforming in JCAS Systems

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Abstract—Hybrid beamforming (HBF) design for joint communication and sensing (JCAS) often results in high computational complexity. Deep unfolded HBF can alleviate this while preserving strong performance. In this work, we propose a knowledge distillation-aided unfolding framework for efficient HBF-JCAS design. Specifically, we design an accelerated unrolled algorithm by transferring learned knowledge from a high-capacity teacher to a lightweight student; both are model-based deep unfolding architectures developed based on the projected gradient ascent (PGA) framework. The teacher helps the student’s learning with teacher-guided initialization (TGI) and layer-wise relational distillation (LRD). While the former accelerates convergence by good initialization, the latter preserves convergence behaviour by distilling structural relations across unfolded layers. Simulation demonstrates that TGI accelerates the unfolded algorithm prominently, and LRD substantially enhances convergence stability. The resulting student can approach the JCAS performance achieved by the teacher with a complexity reduction of 75%.

Index Terms—Hybrid beamforming, deep unfolding, knowledge distillation, joint communication and sensing.

I. INTRODUCTION

Hybrid beamforming (HBF) has become a defining feature of emerging mmWave joint communication and sensing (JCAS) systems, where large antenna arrays provide high spatial resolution while maintaining affordable hardware cost and power consumption compared to fully digital architectures [1], [2]. However, designing hybrid beamformers is fundamentally constrained by the coupled analog–digital architecture, resulting in an inherently non-convex optimization problem under practical hardware constraints [3], [4]. Existing HBF design methods therefore rely on computationally intensive iterative algorithms, limiting their practical applications.

Deep unfolding has emerged as a powerful paradigm for HBF design by transforming iterative optimization algorithms into trainable deep neural networks (DNNs) that preserve the underlying optimization structure [5], [6]. Compared with black-box DNNs, deep-unfolded architectures offer improved interpretability, parameter efficiency, and convergence behaviour [7]–[9]. Nevertheless, sophisticated HBF designs for mmWave massive MIMO JCAS systems often require a large number of unfolded layers, resulting in substantial inference latency and computational complexity [10]. For instance, the deep-unfolded JCAS HBF framework in [11] achieved strong JCAS performance by learning iteration-dependent step sizes. However, it requires up to 120 outer unfolding layers together with multiple inner layers for updating the digital and analog precoders, respectively, leading to considerable complexity.

In this regard, knowledge distillation (KD) offers a promising approach to reducing model complexity while maintaining satisfactory performance. KD enables the training of a compact student model by transferring knowledge from a high-capacity teacher model. Recent studies on beam prediction [12]–[14] and semantic communications [15] have demonstrated that KD can effectively produce low-complexity networks with high performance. Moreover, its potential further extends to CSI feedback compression [16], cross-modal beam prediction [17], and collaborative distributed sensing [18], highlighting the versatility of KD as a model compression framework across various wireless applications.

In this work, we aim to combine the advantages of deep unfolding and knowledge distillation (KD) to develop low-complexity HBF designs for mmWave massive MIMO JCAS systems. Achieving this objective presents several challenges. First, most existing KD methods are tailored to classification tasks and are therefore not directly applicable to regression problems such as HBF design. Second, unlike conventional black-box DNNs, deep-unfolded architectures do not naturally provide the latent feature representations exploited by regression-oriented KD methods, such as relational knowledge distillation (RKD). Furthermore, existing HBF research has primarily focused on architectural modifications to reduce computational complexity [19], while leveraging KD, particularly RKD, to compress deep-unfolded HBF networks remains largely unexplored.

To address these challenges, we propose an accelerated deep-unfolded HBF framework that transfers knowledge from a high-capacity teacher to a lightweight student via two novel techniques: *teacher-guided initialization (TGI)* and *layer-wise relational distillation (LRD)*. Specifically, TGI transfers the teacher’s learned knowledge to initialize the student in a favorable optimization region, mitigating the performance degradation caused by reducing the unfolding depth during distillation. Meanwhile, LRD aligns the relational dynamics of the teacher and student across iterations, enabling the student to inherit the teacher’s convergence behaviour without requiring one-to-one layer correspondence or explicit intermediate supervision. As a result, the proposed framework allows a significantly shallower unfolded network to closely approximate the convergence behaviour of a much deeper teacher model, retaining approximately 97% of the teacher’s objective value while reducing complexity by 75%.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

We consider a mmWave JCAS system, where a base station (BS) equipped with N_t transmit antennas and N_{RF} RF chains simultaneously serves K single-antenna users and

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illuminates a sensing target using a fully connected HBF architecture. The transmit beamformer consists of an analog precoder $\mathbf{V} \in \mathbb{C}^{N \times N_{\text{RF}}}$ and a digital baseband precoder $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_K] \in \mathbb{C}^{N_{\text{RF}} \times K}$, where $\mathbf{f}_k \in \mathbb{C}^{N_{\text{RF}}}$ is the precoding vector for user k . The analog precoder satisfies the unit modulus constraint $|\mathbf{V}_{nm}| = 1, \forall n, m$, where $\mathbf{V}_{nm}$ denotes the (n, m) -th entry of $\mathbf{V}$. The total transmit power constraint is given by $\|\mathbf{V}\mathbf{F}\|_{\text{F}}^2 = P$, where P is the maximum power budget at the BS. Let $\mathbf{h}_k \in \mathbb{C}^{N_t}$ be the channel vector from the BS to user k . The received signal at user k is given by

$$y_k = \mathbf{h}_k^H \mathbf{V} \mathbf{f}_k d_k + \sum_{u \neq k} \mathbf{h}_k^H \mathbf{V} \mathbf{f}_u d_u + n_k, \quad (1)$$

where d_k is the information symbol satisfying $\mathbb{E}[|d_k|^2] = 1$, and $n_k \sim \mathcal{CN}(0, \sigma^2)$ is the additive white Gaussian noise (AWGN) with power σ^2 .

B. Problem Formulation

Our primary objective is to reduce the computational complexity of deep-unfolded HBF architectures by reducing the required number of unfolding layers. To this end, we build upon the deep-unfolded HBF framework in [11] and incorporate the proposed KD mechanisms to accelerate convergence. We adopt the communication sum rate R and the beam pattern mismatch τ as the performance metrics for the communication and sensing subsystems, respectively. Specifically, R and τ are defined as

$$R(\mathbf{V}, \mathbf{F}) = \sum_{k=1}^K \log_2 \left(1 + \frac{|\mathbf{h}_k^H \mathbf{V} \mathbf{f}_k|^2}{\sum_{u \neq k} |\mathbf{h}_k^H \mathbf{V} \mathbf{f}_u|^2 + \sigma^2} \right). \quad (2)$$

$$\tau(\mathbf{V}, \mathbf{F}) = \|\mathbf{V} \mathbf{F} \mathbf{F}^H \mathbf{V}^H - \mathbf{Q}\|_{\text{F}}^2, \quad (3)$$

where $\mathbf{Q} \in \mathbb{C}^{N_t \times K}$ denotes the desired transmit covariance matrix, obtained by minimizing the mismatch between the ideal and synthesized beam patterns [11], [20]. The detailed optimization for $\mathbf{Q}$ is omitted due to space limitations.

The JCAS HBF design is formulated as

$$\max_{\mathbf{V}, \mathbf{F}} R(\mathbf{V}, \mathbf{F}) - \lambda \tau(\mathbf{V}, \mathbf{F}) \quad (4a)$$

$$\text{s.t. } |\mathbf{V}_{nm}| = 1, \quad \forall n, m, \quad (4b)$$

$$\|\mathbf{V} \mathbf{F}\|_{\text{F}}^2 = P, \quad (4c)$$

where $\lambda \in (0, 1)$ controls the communication-sensing tradeoff by weighting the radar beam-alignment error relative to the communication sum rate. In [11], this problem is solved using a combination of alternating optimization (AO) and PGA, which is subsequently unfolded into a trainable DNN with a nested architecture, where the outer unfolding layers update the digital precoder $\mathbf{F}$, while the inner layers update the analog precoder $\mathbf{V}$. This approach significantly improves the communication-sensing performance tradeoff while reducing computational complexity by approximately 65% compared with the conventional AO-PGA algorithm. Nevertheless, it still requires a large number of unfolding layers, resulting in considerable inference complexity. To further reduce the computational complexity while preserving the performance of the deep-unfolded PGA framework, we propose the KD-aided unfolding framework described below.

III. PROPOSED DISTILLATION FRAMEWORK

The main idea of the proposed KD-based deep-unfolding framework is to first train a high-performing model, referred to as the teacher, and then use it to guide the training of a lightweight student model for low-complexity online inference. We first present the general deep-unfolding architecture of the teacher, which also serves as the basis for the student architecture with modified initialization. We then describe how the student model is built by distilling the knowledge of the teacher using the proposed KD framework.

A. Teacher Model

Unfolded Teacher Model: The teacher model follows the deep-unfolded PGA architecture proposed in [11], which consists of nested layers. Specifically, the outer layers update the digital precoder $\mathbf{F}$, while the inner layers update the analog precoder $\mathbf{V}$. Let $\alpha^{(i)}$ denote the learnable step size for updating $\mathbf{F}$ at the i -th outer iteration, and let $\beta^{(j,i)}$ denote the learnable step size for updating $\mathbf{V}$ at the j -th inner iteration of the i -th outer iteration, $i = 1, \dots, I, j = 1, \dots, J$ where I and J are the numbers of outer and inner layers. Furthermore, let $\nabla_{\mathbf{F}} R$, $\nabla_{\mathbf{F}} \tau$, $\nabla_{\mathbf{V}} R$, and $\nabla_{\mathbf{V}} \tau$ denote the gradients of the sum rate R and beam pattern error τ with respect to $\mathbf{F}$ and $\mathbf{V}$, respectively.

At the j -th inner layer of the i -th outer layer, the analog precoder is updated according to

$$\mathbf{V}^{(j+1,i)} = \mathbf{V}^{(j,i)} + \beta^{(j,i)} (\nabla_{\mathbf{V}} R - \lambda \nabla_{\mathbf{V}} \tau) \Big|_{\mathbf{V}=\mathbf{V}^{(j,i)}}. \quad (5)$$

After completing all J inner updates, the resulting analog precoder is projected onto the unit-modulus manifold:

$$\left[\mathbf{V}^{(i)} \right]_{nm} = \frac{[\mathbf{V}^{(J,i)}]_{nm}}{[\mathbf{V}^{(J,i)}]_{nm}}, \quad (6)$$

where $\mathbf{V}^{(i)} \triangleq \mathbf{V}^{(J,i)}$ after the unit-modulus projection. The digital precoder is then updated in the i -th outer layer as

$$\mathbf{F}^{(i+1)} = \mathbf{F}^{(i)} + \alpha^{(i)} (\nabla_{\mathbf{F}} R - \lambda \nabla_{\mathbf{F}} \tau) \Big|_{\mathbf{F}=\mathbf{F}^{(i)}}, \quad (7)$$

followed by power normalization:

$$\mathbf{F}^{(i+1)} = \frac{\sqrt{P_t} \mathbf{F}^{(i+1)}}{\|\mathbf{V}^{(i)} \mathbf{F}^{(i+1)}\|_{\text{F}}}. \quad (8)$$

In (5) and (7), $\nabla_{\mathbf{F}} R$, $\nabla_{\mathbf{F}} \tau$, $\nabla_{\mathbf{V}} R$, and $\nabla_{\mathbf{V}} \tau$ are given in [11], which are omitted here due to the limited space.

Training of Teacher Model: The teacher is trained offline by minimizing the loss function

$$\mathcal{L}_{\text{task}} = -R(\mathbf{V}^{(I)}, \mathbf{F}^{(I)}) + \lambda \tau(\mathbf{V}^{(I)}, \mathbf{F}^{(I)}). \quad (9)$$

In this deep-unfolded DNN architecture, the trainable parameters consist of the step sizes $\{\alpha^{(i)}\}_{i=1}^I$ and $\{\beta^{(j,i)}\}_{(i,j)=(1,1)}^{I,J}$ across all outer and inner layers. These parameters are optimized through unsupervised training to minimize the loss function, which is equivalent to maximizing the underlying objective function.

B. Proposed Student Model

The teacher model employs a large number of outer and inner layers to achieve high performance, resulting in substantial computational complexity. To reduce the inference complexity, we propose distilling the knowledge of the teacher model into

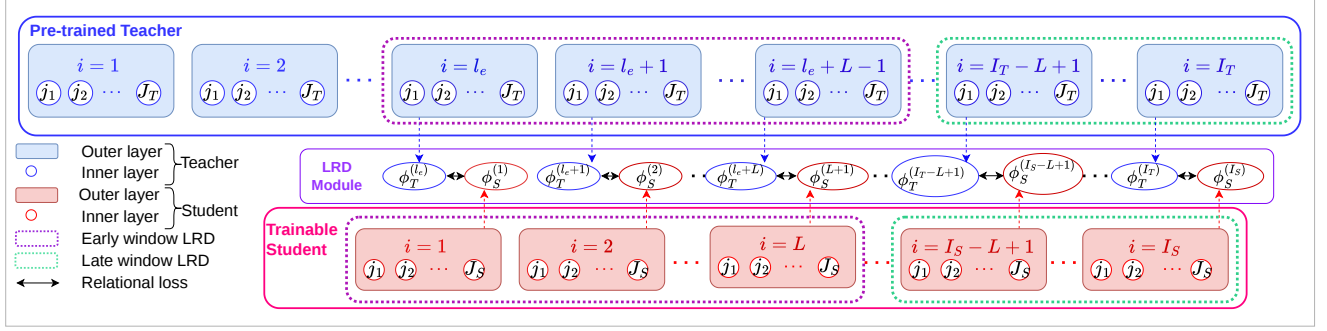


Fig. 1. Illustration of LRD method.

a lightweight student model with significantly fewer layers. The student model is trained using two complementary techniques, namely TGI and LRD, which are described in detail below. For notational simplicity, we use the same notation for the student model as for the teacher model, while appending the subscript S to distinguish the corresponding parameters.

1) Teacher-Guided Initialization

Unlike KD in conventional black-box DNNs, where the teacher's learned weights and biases cannot be directly transferred to the student because of the substantial architectural mismatch, deep-unfolding networks learn algorithm-specific parameters, namely the step sizes, whose physical interpretation is independent of the network depth. Consequently, the well-optimized step sizes learned by the teacher can serve as informative initializations for the student. This observation forms the main motivation behind the proposed TGI.

Specifically, TGI initializes the student step sizes $\{\alpha_s^{(i)}\}$ and $\{\beta_s^{(j,i)}\}$ from the teacher's learned step sizes. However, since the teacher and student generally can employ different numbers of outer and inner layers, i.e., $I \neq I_S$ and/or $J \neq J_S$, the teacher parameters cannot be directly transferred. To address this mismatch, we propose a dimension shrinkage and alignment procedure that compresses the teacher's step-size sequences and maps them to the student architecture while preserving their optimization characteristics.

Let $r = \frac{J}{J_S}$ be the layer reduction factor from the teacher to student. For simplicity, we assume r is an integer. Let $\tilde{\beta}^{(j,i)}$ denote the average step size of each group of r consecutive layers of the teacher, i.e.,

$$\tilde{\beta}^{(j,i)} = \frac{1}{r} \sum_{j'=1}^r \beta^{(jr-j'+1,i)}, \quad \forall i. \quad (10)$$

Then we perform dimension alignment by averaging over all I step sizes of outer layers according to

$$\tilde{\alpha}^{(i)} = \frac{1}{r'} \sum_{i'=1}^{r'} \alpha^{(ir'-i'+1)}, \quad \forall i. \quad (11)$$

[Nhan: Please explain here why you need to average over outer iterations. Then, you should give an example here to make this clear.]

This TGI provides a structured initialization for the student prior to training. Our simulation results show that TGI significantly improves the convergence of the unfolded student model, as will be shown in Section IV.

2) Layer-Wise Relational Distillation

The proposed LRD framework introduces relational supervision over selected iteration steps, as illustrated in Fig. 1. In standard RKD, the final output of the network serves as the embedding to compare relational structure between teacher and student. In contrast, deep-unfolded beamforming produces a sequence of beamformer estimates, one at each unfolding iteration, rather than a single final embedding. LRD exploits this structure by treating the hybrid beamformer at each outer iteration as the embedding, and supervising how the student's beamformer sequence relates to the teacher's at two critical windows: early in training where the objective improves most rapidly, and late where the solution stabilizes near convergence. Within a selected window, the relational comparison is carried out independently at every paired outer layer, with each student layer matched to the corresponding teacher layer as depicted in Fig. 1, ensuring the student learns not only the final solution but also the path taken to reach it.

Specifically, let L be the length of the selected iteration window used for supervising the student model and let l_e be the teacher iteration that marks the beginning of the first window. The *early window* matches student iterates $\{\mathbf{V}_S^{(i)}, \mathbf{F}_S^{(i)}\}_{i=1}^L$ to teacher iterates $\{\mathbf{V}_T^{(i)}, \mathbf{F}_T^{(i)}\}_{i=l_e}^{l_e+L-1}$, enabling rapid early-convergence dynamics. The *late window* matches student iterates $\{\mathbf{V}_S^{(i)}, \mathbf{F}_S^{(i)}\}_{i=I_S-L+1}^{I_S}$ to teacher iterates $\{\mathbf{V}_T^{(i)}, \mathbf{F}_T^{(i)}\}_{i=I_T-L+1}^{I_T}$, encouraging the student to approach the teacher's converged solution. Define $\text{vec}_{\mathbb{R}}(\mathbf{X}) \triangleq [\text{Re}(\text{vec}(\mathbf{X}))^T, \text{Im}(\text{vec}(\mathbf{X}))^T]^T \in \mathbb{R}^{2NM}$ as the operator that vectorizes a complex matrix $\mathbf{X} \in \mathbb{C}^{N \times M}$ into a real-valued vector by stacking its real and imaginary parts. The effective beamformer at the i -th iteration within a window is represented by $\phi_g^{(i,b)} = \text{vec}_{\mathbb{R}}(\mathbf{V}_g^{(i,b)} \mathbf{F}_g^{(i,b)}) \in \mathbb{R}^{2NK}$, where $b = 1, \dots, B$ is the sample index in a mini-batch of size B , and $\mathbf{V}_g^{(i,b)} \mathbf{F}_g^{(i,b)}$ denotes the effective beamformer, combining the analog and digital precoders, corresponding to the b -th sample at the i -th iteration. For each sample pair (b, b') and triplet (b, b', b'') , the distance-based and angle-based relational potentials are defined as

$$\psi_g^{\text{dist}} = \frac{\|\phi_g^{(i,b)} - \phi_g^{(i,b')}\|_2}{\mu_d}, \quad (12)$$

$$\psi_g^{\text{angl}} = \cos \langle \phi_g^{(i,b')} - \phi_g^{(i,b)}, \phi_g^{(i,b'')} - \phi_g^{(i,b)} \rangle, \quad (13)$$

where μ_d denotes the mean pairwise Euclidean distance over all sample pairs in the mini-batch. The discrepancy between

the teacher and student relational potentials is measured using the Huber loss $w(x, y)$ [21]. The distance and angle losses are then computed as

$$\mathcal{L}_{\text{dist}}^{(i)} = \binom{B}{2}^{-1} \sum_{\mathcal{B}^2} w(\psi^{\text{dist}}, \psi_s^{\text{dist}}),$$

and

$$\mathcal{L}_{\text{angle}}^{(i)} = \binom{B}{3}^{-1} \sum_{\mathcal{B}^3} w(\psi^{\text{angl}}, \psi_s^{\text{angl}}),$$

where $\binom{B}{p}$ denotes the combinatorial number. Here, $\mathcal{B}^2$ and $\mathcal{B}^3$ denote the sets of all sample pairs and triplets in the mini-batch, respectively. The cross-iteration relational loss within each window is given by

$$\mathcal{L}_{\text{CI}} = \sum_{i=1}^L \gamma_i \left(\lambda_d \mathcal{L}_{\text{dist}}^{(i)} + \lambda_a \mathcal{L}_{\text{angle}}^{(i)} \right), \quad (14)$$

where $\gamma_i = \log(i)$ is a depth-dependent weight assigning stronger supervision to later iterations; λ_d and λ_a are weighting coefficients balancing the distance and angle losses. To further supervise the student beyond the two windows, we introduce a log-weighted deep supervision loss

$$\mathcal{L}_{\text{log}} = \sum_{i=1}^{I_s} \gamma_i \left\| \mathcal{J}^{(I)} - \mathcal{J}_s^{(i)} \right\|^2, \quad (15)$$

where $\mathcal{J}_g^{(i)} = R^{(i)} - \lambda \tau^{(i)}$ denotes the JCAS objective at each outer layer, and $\mathcal{J}^{(I)}$ is the teacher's final objective serving as a fixed reference target. The logarithmic weighting assigns negligible supervision to early, unreliable layers and progressively stronger supervision as the student approaches convergence, complementing the window-based relational losses. The complete student training objective is given by

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{log}} \mathcal{L}_{\text{log}} + \mathcal{L}_{\text{CI}}^{\text{early}} + \lambda_{\text{late}} \mathcal{L}_{\text{CI}}^{\text{late}}, \quad (16)$$

where $\lambda_{\text{late}} > 0$ and $\lambda_{\text{log}} > 0$ are adjustable hyperparameters. The overall training procedure is summarized in Algorithm 1.

C. Complexity Analysis

As the model training can be performed offline, we only evaluate the model's inference complexity. The dominant computational cost arises from evaluating the analog precoder gradients in (7). Computing $\nabla_{\mathbf{V}} R$ and $\nabla_{\mathbf{V}} \tau$ involves matrix products of the form $\mathbf{H}_k \mathbf{F} \mathbf{V}$ and $\mathbf{V} \mathbf{F} \mathbf{V}^H \mathbf{H}_k$, where $\mathbf{H}_k = \mathbf{h}_k \mathbf{h}_k^H \in \mathbb{C}^{N \times N}$ and $\mathbf{F} \mathbf{F}^H \in \mathbb{C}^{N_{\text{RF}} \times N_{\text{RF}}}$, each costing $\mathcal{O}(NN_{\text{RF}}K)$ and $\mathcal{O}(N^2K)$ respectively. Since digital precoder gradient updates and projection operations incur no greater cost, one outer iteration of J_s inner steps costs $\mathcal{O}(J_s KN^2)$. Over I_s outer iterations, the deployed student incurs $\mathcal{C}_s = \mathcal{O}(I_s J_s KN^2)$, whereas the teacher incurs $\mathcal{C} = \mathcal{O}(IJKN^2)$. Since both networks share identical per-step calculations, their complexity ratio depends solely on unrolled depth and is therefore $\frac{\mathcal{C}_s}{\mathcal{C}} = \frac{I_s J_s}{IJ} = q$. The student thus achieves a $q \times$ complexity reduction. Since the student unfolded depth is adjustable, the proposed framework offers a flexible trade-off between inference cost and beamforming performance.

Algorithm 1: LRD Training of the Student Model

Input: Training set $\mathcal{D}_{\text{tr}}; \{\alpha^{(i)}, \beta^{(j,i)}\}; \{\alpha_s^{(i)}, \beta_s^{(j,i)}\};$
hyperparameters $\{L, l_e, \lambda_d, \lambda_a, \lambda_{\text{log}}, \lambda_{\text{late}}\};$ epochs E .

Output: Optimized precoders $\mathbf{F}^{(I_s)}$ and $\mathbf{V}^{(I_s)}$ obtained using $\{\alpha_s^*, \beta_s^*\}$.

- 1 Initialize precoders following [11]
- 2 **for** $\hat{e} = 1, \dots, E$ **do**
- 3 **for** each mini-batch $\mathcal{B} \subseteq \mathcal{D}_{\text{tr}}, |\mathcal{B}| \geq 3$ **do**
- 4 Obtain teacher precoder iterates $\{\mathbf{F}^{(i)}, \mathbf{V}^{(i)}\}$ via $\{\alpha^{(i)}, \beta^{(j,i)}\}$ and extract window iterates at $i \in [l_e, l_e + L - 1] \cup [I - L + 1, I]$.
- 5 Obtain student precoder iterates $\{\mathbf{F}_s^{(i)}, \mathbf{V}_s^{(i)}\}$ via $\{\alpha_s^{(i)}, \beta_s^{(j,i)}\}$ and extract window iterates at $i \in [1, L] \cup [I_s - L + 1, I_s]$.
- 6 Compute $\mathcal{L}_{\text{task}} = -R(\mathbf{V}_s^{(I_s)}, \mathbf{F}_s^{(I_s)}) + \lambda \tau(\mathbf{V}_s^{(I_s)}, \mathbf{F}_s^{(I_s)})$.
- 7 Compute $\mathcal{L}_{\text{log}}, \mathcal{L}_{\text{CI}}, \mathcal{L}_{\text{total}}$ via (15), (14); (16).
- 8 Update $\{\alpha_s^{(i)}, \beta_s^{(j,i)}\}$ via backpropagation and Adam optimizer.
- 9 **end**
- 10 **end**
- 11 **return** $\mathbf{F}^{(I_s)}, \mathbf{V}^{(I_s)}$ using $\{\alpha_s^*, \beta_s^*\}$.

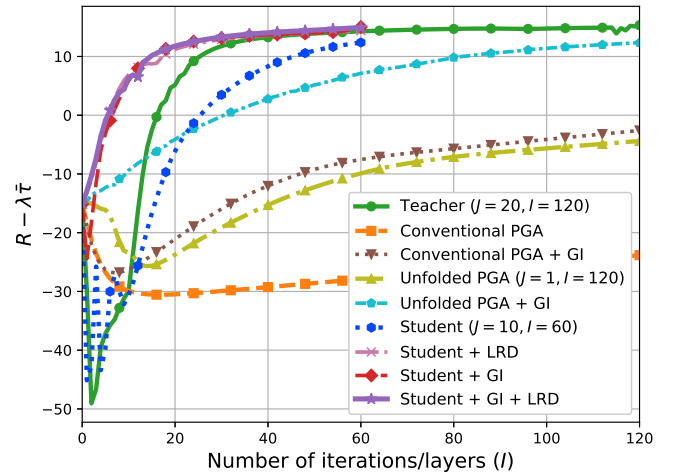


Fig. 2. Convergence of JCAS objective $R - \lambda \tau$.

IV. NUMERICAL RESULTS

In the simulation, we consider $N = 64$ transmit antennas, $N_{\text{RF}} = 4$ RF chains, and $K = 4$ single-antenna users. Training dataset consists of 420 channel realizations spanning signal-to-noise ratios (SNRs) from 0 to 12 dB in 2 dB increments, where SNR is defined as $\text{SNR} = \frac{P}{\sigma^2}$. Simulation results are averaged over 10^2 independent test channel realizations. The teacher uses $(J, I) = (20, 120)$ and is pretrained offline. The proposed student uses $(J_s, I_s) = (10, 60)$ and is trained for 30 epochs using the Adam optimizer with a learning rate of 10^{-3} and a batch size of 10. Unless otherwise stated, the LRD hyperparameters are fixed to $L = 15$, $l_e = 20$, $\lambda_d = 25$, $\lambda_a = 50$, $\lambda_{\text{late}} = 0.8$, and $\lambda_{\text{log}} = 0.0001$. Table I summarizes the objective, sum rate and beam-alignment error achieved at the corresponding iteration budgets $(J \times I)$ for all evaluated models at $\text{SNR} = 12\text{dB}$ and $\lambda = 0.3$.

Fig. 2 compares the evolution of the joint communication-sensing objective with the unfolding depth at $\text{SNR} = 12\text{ dB}$.

As expected, reducing the iteration count degrades both convergence speed and objective, indicating that the shallow student without RKD cannot reproduce the optimization quality of the teacher. Incorporating TGI recovers most of the lost performance, with Student+TGI reaching within 2% of the teacher's objective at a $4\times$ reduction in iterations. Combining TGI with LRD yields a slight but comparable performance to TGI alone while further accelerating convergence toward the teacher solution.

Table I further shows that iteration reduction alone is insufficient without a principled acceleration mechanism. Conventional PGA with only 120 iterations yields a negative objective due to poorly controlled beam error. In contrast, Unfolded PGA+TGI improves the objective drastically by 381.3% over unfolded PGA without TGI at the same iteration budget, while retaining over 80% of the teacher's objective at only 5% of its iteration cost. Additionally, Student+TGI and Student+TGI+LRD attain 98.0% and 97.3% of the teacher joint objective, respectively, at 25% of the teacher's inference cost, whereas Student+LRD independently reaches 93.8% of the teacher, confirming the complementary contribution of LRD even without structured initialization. Together, these results indicate that TGI and LRD enable substantially accelerated convergence while preserving near-teacher performance, addressing the complexity-performance tradeoff.

Figs. 3 and 4 compare the communication-sensing tradeoff across SNR. All learned methods achieve similar communication rates, with only minor differences at high SNR. In contrast, above $\text{SNR} = 6\text{dB}$, sensing performance diverges more noticeably across the evaluated methods, with the teacher consistently achieving the lowest beam error. Although Unfolded PGA+TGI and the non-distilled student attain the highest communication rates at high SNR, they exhibit the weakest sensing performance, illustrating the inherent communication-sensing tradeoff. Overall, the proposed TGI and LRD consistently outperform the non-distilled student operating at the same inference budget.

V. CONCLUSION

This paper investigated a novel knowledge distillation approach for deep-unfolded HBF in JCAS systems. To the best of our knowledge, this is the first work that jointly exploits structured initialization transfer and relational distillation for deep-unfolding. The results demonstrate that simply reducing the unfolding depth leads to significant performance loss. In contrast, convergence-aware distillation enables a shallow network to recover the convergence dynamics of a substantially deeper teacher. In particular, the simulation results revealed

TABLE I. Model performance at $\text{SNR} = 12\text{ dB}$, $\lambda = 0.3$.

Model	$R - \lambda\tau$	R [b/s/Hz]	τ	$J \times I$
Teacher	15.32	22.30	23.29	2400
Conventional PGA	-23.83	16.18	133.35	120
Conventional PGA + TGI	-2.59	28.88	104.95	120
Unfolded PGA	-4.38	29.05	111.46	120
Unfolded PGA+TGI	12.33	23.54	37.36	120
Student	12.38	23.28	36.34	600
Student+LRD	14.37	22.31	26.46	600
Student+TGI	15.02	22.33	24.36	600
Student+TGI+LRD	14.91	22.24	24.44	600

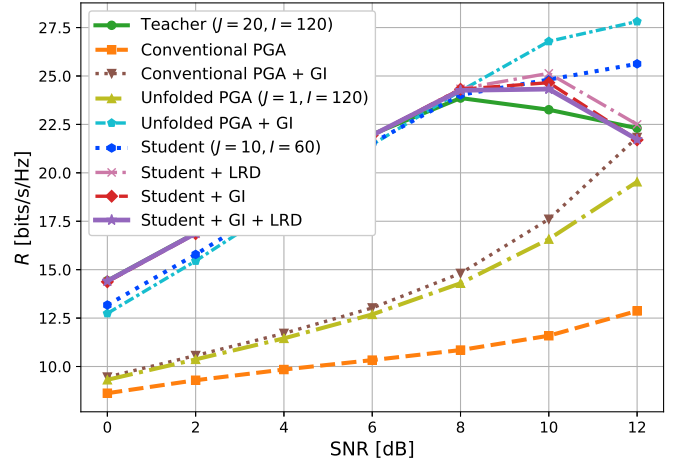


Fig. 3. Sum rate.

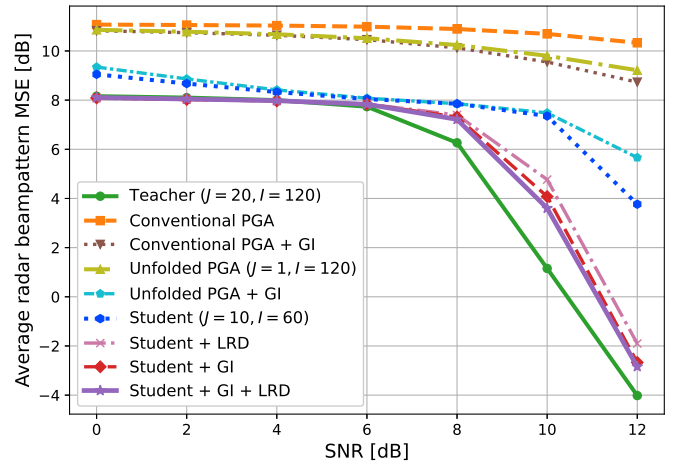


Fig. 4. Mean beam error $\bar{\tau}$.

that TGI is primarily responsible for accelerating convergence by improving optimization efficiency, while LRD helps to recover the attainable solution quality. As a result, the proposed student achieves near-teacher communication-sensing performance at only 25% of the teacher's model cost. The findings further depict that the geometry of an optimization algorithm constitutes a transferable inductive bias that can be extended to broader classes of unfolded optimization methods, which is key to developing lightweight deep-unfolded networks in future wireless systems.

REFERENCES

- [1] S. Lu, F. Liu, Y. Li, K. Zhang, H. Huang, J. Zou, X. Li, Y. Dong, F. Dong, J. Zhu, Y. Xiong, W. Yuan, Y. Cui, and L. Hanzo, "Integrated sensing and communications: Recent advances and ten open challenges," *IEEE Internet Things J.*, vol. 11, no. 11, pp. 19 094–19 120, 2024.
- [2] J. Zhang, M. Liu, J. Tang, and X. Wang, "Deep unfolding learning aided ISAC transceiver design," *IEEE Trans. Wireless Commun.*, vol. 24, no. 10, pp. 8681–8695, 2025.
- [3] N. T. Nguyen, L. V. Nguyen, N. Shlezinger, Y. C. Eldar, A. L. Swindlehurst, and M. Juntti, "Joint communications and sensing hybrid beamforming design via deep unfolding," *IEEE J. Sel. Topics Signal Process.*, vol. 18, no. 5, pp. 901–916, 2024.
- [4] J. Zhang, C. Masouros, F. Liu, Y. Huang, and A. L. Swindlehurst, "Low-complexity joint radar-communication beamforming: From optimization to deep unfolding," *IEEE J. Sel. Topics Signal Process.*, pp. 1–16, 2025, in press.
- [5] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning," *Proc. IEEE*, vol. 111, no. 5, pp. 465–499, 2023.
- [6] S. Deka, K. Deka, N. T. Nguyen, S. Sharma, V. Bhatia, and N. Rajatheva, "Comprehensive review of deep unfolding techniques for next-

- generation wireless communication systems,” *IEEE Internet Things J.*, vol. 13, no. 6, pp. 10379–10406, 2026.
- [7] N. T. Nguyen, M. Ma, O. Lavi, N. Shlezinger, Y. C. Eldar, A. L. Swindlehurst, and M. Juntti, “Deep unfolding hybrid beamforming designs for THz massive MIMO systems,” *IEEE Trans. Signal Process.*, vol. 71, pp. 3788–3804, 2023.
 - [8] A. Chowdhury, G. Verma, A. Swami, and S. Segarra, “Deep graph unfolding for beamforming in MU-MIMO interference networks,” *IEEE Trans. Wireless Commun.*, vol. 23, no. 5, pp. 4889–4903, 2024.
 - [9] B. Xu, J. Zhang, Q. Lin, H. Xiao, Y. Wu, and B. Ai, “Deep unfolding beamforming and power control designs for multi-port matching networks,” *IEEE Trans. Wireless Commun.*, vol. 24, no. 2, pp. 1401–1414, 2025.
 - [10] N. Shlezinger, M. Ma, O. Lavi, N. T. Nguyen, Y. C. Eldar, and M. Juntti, “Artificial intelligence-empowered hybrid MIMO beamforming,” *IEEE Veh. Technol. Mag.*, vol. 19, no. 3, pp. 58–67, 2024.
 - [11] N. T. Nguyen, L. V. Nguyen, N. Shlezinger, Y. C. Eldar, A. L. Swindlehurst, and M. Juntti, “Joint communications and sensing hybrid beamforming design via deep unfolding,” *IEEE J. Sel. Topics Signal Process.*, vol. 18, no. 5, pp. 901–916, 2024.
 - [12] M. Ma, N. T. Nguyen, N. Shlezinger, Y. C. Eldar, A. L. Swindlehurst, and M. Juntti, “Knowledge distillation for sensing-assisted long-term beam tracking in mmwave communications,” *IEEE Trans. Wireless Commun.*, 2026 (Early access).
 - [13] M. Ma, I. Welgamage, A. Alkhateeb, A. L. Swindlehurst, M. Juntti, and N. T. Nguyen, “Knowledge distillation for lightweight multimodal sensing-aided mmwave beam tracking,” *arXiv preprint arXiv:2604.16708*, 2026.
 - [14] S. Tavakolian, N. T. Nguyen, A. Alkhateeb, and M. Juntti, “Knowledge distillation for mmWave beam prediction using sub-6 GHz channels,” in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2026, pp. 22 642–22 646.
 - [15] C. Liu, Y. Zhou, Y. Chen, and S.-H. Yang, “Knowledge distillation-based semantic communications for multiple users,” *IEEE Trans. Wireless Commun.*, vol. 23, no. 7, pp. 7000–7012, 2024.
 - [16] Y. Cui, X. Li, W. Zhang, J. Zhang, and M. Matthaiou, “Lightweight neural network with knowledge distillation for CSI feedback,” *IEEE Trans. Commun.*, vol. 72, no. 8, pp. 4917–4929, 2024.
 - [17] J. Yang, S. Chang, S. Xu, L. Zhou, S. Huang, and Z. Hong, “Cross-modal knowledge distillation for efficient radar-only beam prediction in mmWave communications,” in *arXiv preprint arXiv:2511.19947*, 2025.
 - [18] N. T. Nguyen, M. Ma, N. Shlezinger, J. Choi, Y. C. Eldar, A. L. Swindlehurst, and M. Juntti, “Knowledge distillation for collaborative learning in distributed communications and sensing,” *arXiv preprint arXiv:2603.16116*, 2026.
 - [19] A. Milstein, T. Yablonka, and N. Shlezinger, “Learned approximated optimization for rapid low-complexity hybrid beamforming design,” in *Proc. IEEE ICASSP*, 2025.
 - [20] F. Liu, C. Masouros, A. Li, H. Sun, and L. Hanzo, “MU-MIMO communications with MIMO radar: From co-existence to joint transmission,” *IEEE Trans. Wireless Commun.*, vol. 17, no. 4, pp. 2755–2770, 2018.
 - [21] W. Park, D. Kim, Y. Lu, and M. Cho, “Relational knowledge distillation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019.